



University Mohamed Boudiaf of M'sila

Department of Computer Science

Master 1 – Artificial Intelligence

Module: FDTA — Fouille de Données, Techniques et Applications

Lab Report – TP N°3

Association Rules with R

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Academic Year: 2025 / 2026

Objective

This lab applies the Apriori algorithm to mine association rules from the built-in Titanic dataset in R. The goal is to discover meaningful patterns linking passenger attributes — such as class, gender, and age — to survival outcomes, using the arules package with configurable support and confidence thresholds.

Step 1: Installing and Loading the arules Package

The arules package is installed from CRAN using `install.packages()`, which downloads the compiled binary and places it in the local R library. Once installed, `library(arules)` loads the package into the session and makes its functions available for use.

```
install.packages("arules")
library(arules)

> install.packages("arules")

WARNING: Rtools is required to build R packages but is not currently installed. Please download and install
the appropriate version of Rtools before proceeding:

https://cran.rstudio.com/bin/windows/Rtools/

Installing package into 'C:/Users/Younes/AppData/Local/R/win-library/4.5'
(as 'lib' is unspecified)

trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/arules_1.7.14.zip'

Content type 'application/zip' length 2635777 bytes (2.5 MB)
downloaded 2.5 MB

package 'arules' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
  C:\Users\Younes\AppData\Local\Temp\RtmpkjrM4N\downloaded_packages

> library(arules)

Loading required package: Matrix

Attaching package: 'arules'

The following objects are masked from 'package:base':

  abbreviate, write

Warning message:
package 'arules' was built under R version 4.5.3
```

Step 2: Exploring the Titanic Dataset Structure

The built-in Titanic dataset is a 4-dimensional contingency table. Calling `str(Titanic)` reveals its dimensions and the categorical levels for Class, Sex, Age, and Survived. This inspection confirms the dataset is a numeric table indexed by four factors, not a flat data frame.

```
str(Titanic)

> str(Titanic)
'table' num [1:4, 1:2, 1:2, 1:2] 0 0 35 0 0 0 17 0 118 154 ...
- attr(*, "dimnames")=List of 4
 ..$ Class      : chr [1:4] "1st" "2nd" "3rd" "Crew"
 ..$ Sex        : chr [1:2] "Male" "Female"
 ..$ Age        : chr [1:2] "Child" "Adult"
 ..$ Survived: chr [1:2] "No" "Yes"
```

Step 3: Converting Titanic to a Data Frame

The `as.data.frame()` function flattens the multi-dimensional table into a tidy tabular structure with one row per combination of Class, Sex, Age, and Survived, together with a Freq column indicating how many passengers belong to each group. The result is stored in `df` and viewed both with `head()` and the RStudio data viewer.

```
df <- as.data.frame(Titanic)
head(df)
```

```
> df <- as.data.frame(Titanic)
```



	Class	Sex	Age	Survived	Freq
1	1st	Male	Child	No	0
2	2nd	Male	Child	No	0
3	3rd	Male	Child	No	35
4	Crew	Male	Child	No	0
5	1st	Female	Child	No	0
6	2nd	Female	Child	No	0
7	3rd	Female	Child	No	17
8	Crew	Female	Child	No	0
9	1st	Male	Adult	No	118
10	2nd	Male	Adult	No	154

```
> head(df)
  Class Sex Age Survived Freq
1  1st Male Child      No    0
2  2nd Male Child      No    0
3  3rd Male Child      No   35
4 Crew Male Child      No    0
5  1st Female Child      No    0
6  2nd Female Child      No    0
```

Step 4: Building the Raw Transaction Dataset (titanic.raw)

Since the Apriori algorithm requires one transaction per individual, each aggregated row in `df` is expanded by repeating its category values `Freq` times. The loop iterates over the four attribute columns, binding repeated character vectors column-by-column into `titanic.raw` before converting it to a data frame and assigning proper column names.

```
titanic.raw <- NULL
for(i in 1:4) {
  titanic.raw <- cbind(titanic.raw, rep(as.character(df[,i]), df$Freq))
}
titanic.raw <- as.data.frame(titanic.raw)
names(titanic.raw) <- names(df)[1:4]
```

```
> titanic.raw <- NULL
```

Values	
titanic.raw	NULL

```
> for(i in 1:4) {
+   titanic.raw <- cbind(titanic.raw, rep(as.character(df[,i]), df$F
+ }
```

titanic.raw	chr [1:2201, 1:4] "3rd" "3rd" "3rd" "... 
-------------	--

Values	
i	4L

```
> titanic.raw <- as.data.frame(titanic.raw)
```

Step 5: Inspecting the Expanded Dataset

After expansion, `titanic.raw` contains 2,201 observations — one per passenger — across 4 character variables. The RStudio viewer shows each row as an individual record, `dim()` confirms the 2201×4 shape, and `str()` validates that all columns are character type ready for transaction conversion.

```
dim(titanic.raw)
str(titanic.raw)
head(titanic.raw)
summary(titanic.raw)
```

Filter				
	V1	V2	V3	V4
1	3rd	Male	Child	No
2	3rd	Male	Child	No
3	3rd	Male	Child	No
4	3rd	Male	Child	No
5	3rd	Male	Child	No
6	3rd	Male	Child	No
7	3rd	Male	Child	No
8	3rd	Male	Child	No
9	3rd	Male	Child	No
10	3rd	Male	Child	No

```
> names(titanic.raw) <- names(df)[1:4]
```

titanic.raw	2201 obs. of 4 variables 
\$ Class	: chr "3rd" "3rd" "3rd" "3rd" ...
\$ Sex	: chr "Male" "Male" "Male" "Male" ...
\$ Age	: chr "Child" "Child" "Child" "Child" ...
\$ Survived	: chr "No" "No" "No" "No" ...


```

> dim(titanic.raw)
[1] 2201    4

> str(titanic.raw)
'data.frame':   2201 obs. of  4 variables:
 $ Class      : chr  "3rd" "3rd" "3rd" "3rd" ...
 $ Sex        : chr  "Male" "Male" "Male" "Male" ...
 $ Age        : chr  "Child" "Child" "Child" "Child" ...
 $ Survived   : chr  "No" "No" "No" "No" ...

> head(titanic.raw)
  Class Sex Age Survived
1  3rd Male Child      No
2  3rd Male Child      No
3  3rd Male Child      No
4  3rd Male Child      No
5  3rd Male Child      No
6  3rd Male Child      No

> summary(titanic.raw)
      Class                Sex                Age                Survived
Length:2201      Length:2201      Length:2201      Length:2201
Class :character  Class :character  Class :character  Class :cha
Mode  :character  Mode  :character  Mode  :character  Mode  :cha

```

Step 6: Applying the Apriori Algorithm (Default Parameters)

The `apriori()` function is called with default thresholds: minimum support of 10%, minimum confidence of 80%, and a maximum rule length of 10. The algorithm processes 2,201 transactions across 10 items, scans candidate itemsets of sizes 1 through 4, and outputs exactly 27 association rules stored in `rules.all`.

```

rules.all <- apriori(titanic.raw)
rules.all
inspect(rules.all)

```

```

> rules.all <- apriori(titanic.raw)
Apriori

Parameter specification:
 confidence minval  smax  arem  aval originalSupport  maxtime support minlen maxlen target  ext
  0.8       0.1    1 none FALSE             TRUE         5     0.1     1    10  rules TRUE

Algorithmic control:
 filter tree heap memopt load sort verbose
  0.1 TRUE TRUE  FALSE TRUE    2    TRUE

Absolute minimum support count: 220

set item appearances ...[0 item(s)] done [0.00s].
set transactions ...[10 item(s), 2201 transaction(s)] done [0.00s].
sorting and recoding items ... [9 item(s)] done [0.00s].
creating transaction tree ... done [0.00s].
checking subsets of size 1 2 3 4 done [0.00s].
writing ... [27 rule(s)] done [0.00s].
creating S4 object ... done [0.00s].

Warning message:
Column(s) 1, 2, 3, 4 not logical or factor. Applying default discretization (see '? discretizeDF').

```

Name	Type	Value
rules.all	S4 (arules::rules)	S4 object of class rules
lhs	S4 [27 x 10] (arules::itemMatrix)	S4 object of class itemMatrix
data	S4 [10 x 27] (Matrix::ngCMatrix)	S4 object of class ngCMatrix
itemInfo	list [10 x 3] (S3: data.frame)	A data.frame with 10 rows and 3 columns
itemsetInfo	list [0 x 0] (S3: data.frame)	A data.frame with 0 rows and 0 columns
rhs	S4 [27 x 10] (arules::itemMatrix)	S4 object of class itemMatrix
data	S4 [10 x 27] (Matrix::ngCMatrix)	S4 object of class ngCMatrix
itemInfo	list [10 x 3] (S3: data.frame)	A data.frame with 10 rows and 3 columns
itemsetInfo	list [0 x 0] (S3: data.frame)	A data.frame with 0 rows and 0 columns
quality	list [27 x 5] (S3: data.frame)	A data.frame with 27 rows and 5 columns
support	double [27]	0.950 0.119 0.145 0.193 0.285 0.297 ...
confidence	double [27]	0.950 0.916 0.982 0.904 0.888 0.920 ...
coverage	double [27]	1.000 0.129 0.148 0.214 0.321 0.323 ...
lift	double [27]	1.000 0.964 1.033 0.951 0.934 0.968 ...
count	integer [27]	2092 261 319 425 627 654 ...
info	list [5]	List of length 5
data	symbol	`titanic.raw`
ntransactions	integer [1]	2201
support	double [1]	0.1
confidence	double [1]	0.8
call	character [1]	'apriori(data = titanic.raw)'

```
> rules.all
set of 27 rules
```

```
> inspect(rules.all)
  lhs                                     rhs      support  confidence coverage  lift
[1] {}                                     => {Age=Adult} 0.9504771 0.9504771 1.0000000 1.0000000
[2] {Class=2nd}                           => {Age=Adult} 0.1185825 0.9157895 0.1294866 0.9635051
[3] {Class=1st}                           => {Age=Adult} 0.1449341 0.9815385 0.1476602 1.0326798
[4] {Sex=Female}                           => {Age=Adult} 0.1930940 0.9042553 0.2135393 0.9513700
[5] {Class=3rd}                           => {Age=Adult} 0.2848705 0.8881020 0.3207633 0.9343750
[6] {Survived=Yes}                        => {Age=Adult} 0.2971377 0.9198312 0.3230350 0.9677574
[7] {Class=Crew}                          => {Sex=Male} 0.3916402 0.9740113 0.4020900 1.2384742
[8] {Class=Crew}                          => {Age=Adult} 0.4020900 1.0000000 0.4020900 1.0521033
[9] {Survived=No}                         => {Sex=Male} 0.6197183 0.9154362 0.6769650 1.1639949
[10] {Survived=No}                       => {Age=Adult} 0.6533394 0.9651007 0.6769650 1.0153856
[11] {Sex=Male}                           => {Age=Adult} 0.7573830 0.9630272 0.7864607 1.0132040
[12] {Sex=Female, Survived=Yes}           => {Age=Adult} 0.1435711 0.9186047 0.1562926 0.9664669
[13] {Class=3rd, Sex=Male}                 => {Survived=No} 0.1917310 0.8274510 0.2317129 1.2222950
[14] {Class=3rd, Survived=No}              => {Age=Adult} 0.2162653 0.9015152 0.2398910 0.9484870
[15] {Class=3rd, Sex=Male}                 => {Age=Adult} 0.2099046 0.9058824 0.2317129 0.9530818
[16] {Sex=Male, Survived=Yes}              => {Age=Adult} 0.1535666 0.9209809 0.1667424 0.9689670
[17] {Class=Crew, Survived=No}             => {Sex=Male} 0.3044071 0.9955423 0.3057701 1.2658514
[18] {Class=Crew, Survived=No}             => {Age=Adult} 0.3057701 1.0000000 0.3057701 1.0521033
[19] {Class=Crew, Sex=Male}                => {Age=Adult} 0.3916402 1.0000000 0.3916402 1.0521033
[20] {Class=Crew, Age=Adult}                => {Sex=Male} 0.3916402 0.9740113 0.4020900 1.2384742
[21] {Sex=Male, Survived=No}               => {Age=Adult} 0.6038164 0.9743402 0.6197183 1.0251065
[22] {Age=Adult, Survived=No}              => {Sex=Male} 0.6038164 0.9242003 0.6533394 1.1751385
[23] {Class=3rd, Sex=Male, Survived=No}    => {Age=Adult} 0.1758292 0.9170616 0.1917310 0.9648435
[24] {Class=3rd, Age=Adult, Survived=No}    => {Sex=Male} 0.1758292 0.8130252 0.2162653 1.0337773
[25] {Class=3rd, Sex=Male, Age=Adult}       => {Survived=No} 0.1758292 0.8376623 0.2099046 1.2373791
[26] {Class=Crew, Sex=Male, Survived=No}    => {Age=Adult} 0.3044071 1.0000000 0.3044071 1.0521033
[27] {Class=Crew, Age=Adult, Survived=No}  => {Sex=Male} 0.3044071 0.9955423 0.3057701 1.2658514
count
[1] 2092
[2] 261
[3] 319
[4] 425
[5] 627
[6] 654
```

Step 7: Filtering Rules — Targeting Survival Outcomes

To focus the analysis on survival prediction, a second `apriori()` call restricts the right-hand side (rhs) to only "Survived=No" or "Survived=Yes". The minimum support is lowered to 0.5% and minimum length set to 2 to capture less frequent but informative patterns; the resulting 12 rules are rounded and sorted by lift.

```
rules <- apriori(titanic.raw,
  control = list(verbose=F),
  parameter = list(minlen=2, supp=0.005, conf=0.8),
  appearance = list(rhs=c("Survived=No", "Survived=Yes"),
    default="lhs"))
quality(rules) <- round(quality(rules), digits=3)
rules.sorted <- sort(rules, by="lift")
inspect(rules.sorted)

> rules <- apriori(titanic.raw, control = list(verbose=F), parameter = list(minlen=2, supp=0.005, conf=0.8), appearance
= list(rhs=c("Survived=No", "Survived=Yes"), default="lhs"))
```


Name	Type	Value
rules	S4 (arules::rules)	S4 object of class rules
lhs	S4 [12 x 10] (arules::itemMatrix)	S4 object of class itemMatrix
data	S4 [10 x 12] (Matrix::ngCMatrix)	S4 object of class ngCMatrix
itemInfo	list [10 x 3] (S3: data.frame)	A data.frame with 10 rows and 3 columns
itemsetInfo	list [0 x 0] (S3: data.frame)	A data.frame with 0 rows and 0 columns
rhs	S4 [12 x 10] (arules::itemMatrix)	S4 object of class itemMatrix
data	S4 [10 x 12] (Matrix::ngCMatrix)	S4 object of class ngCMatrix
itemInfo	list [10 x 3] (S3: data.frame)	A data.frame with 10 rows and 3 columns
itemsetInfo	list [0 x 0] (S3: data.frame)	A data.frame with 0 rows and 0 columns
quality	list [12 x 5] (S3: data.frame)	A data.frame with 12 rows and 5 columns
support	double [12]	0.01090 0.04225 0.06997 0.06406 0.00909 0.19173 ...
confidence	double [12]	1.000 0.877 0.860 0.972 0.870 0.827 ...
coverage	double [12]	0.0109 0.0482 0.0813 0.0659 0.0104 0.2317 ...
lift	double [12]	3.10 2.72 1.27 3.01 2.69 1.22 ...
count	integer [12]	24 93 154 141 20 422 ...
info	list [5]	List of length 5
data	symbol	`titanic.raw`
ntransactions	integer [1]	2201
support	double [1]	0.005
confidence	double [1]	0.8
call	character [1]	'apriori(data = titanic.raw, parameter = list(minlen = 2, supp = 0.005, conf = 0 ...

```
> quality(rules) <- round(quality(rules), digits=3)
> rules.sorted <- sort(rules, by="lift")
```


Name	Type	Value
rules.sorted	S4 (arules::rules)	S4 object of class rules
lhs	S4 [12 x 10] (arules::itemMatrix)	S4 object of class itemMatrix
data	S4 [10 x 12] (Matrix::ngCMatrix)	S4 object of class ngCMatrix
itemInfo	list [10 x 3] (S3: data.frame)	A data.frame with 10 rows and 3 columns
itemsetInfo	list [0 x 0] (S3: data.frame)	A data.frame with 0 rows and 0 columns
rhs	S4 [12 x 10] (arules::itemMatrix)	S4 object of class itemMatrix
data	S4 [10 x 12] (Matrix::ngCMatrix)	S4 object of class ngCMatrix
itemInfo	list [10 x 3] (S3: data.frame)	A data.frame with 10 rows and 3 columns
itemsetInfo	list [0 x 0] (S3: data.frame)	A data.frame with 0 rows and 0 columns
quality	list [12 x 5] (S3: data.frame)	A data.frame with 12 rows and 5 columns
support	double [12]	0.011 0.006 0.064 0.064 0.042 0.009 ...
confidence	double [12]	1.000 1.000 0.972 0.972 0.877 0.870 ...
coverage	double [12]	0.011 0.006 0.066 0.065 0.048 0.010 ...
lift	double [12]	3.10 3.10 3.01 3.01 2.72 2.69 ...
count	double [12]	24 13 141 140 93 20 ...
info	list [5]	List of length 5
data	symbol	`titanic.raw`
ntransactions	integer [1]	2201
support	double [1]	0.005
confidence	double [1]	0.8
call	character [1]	`apriori(data = titanic.raw, parameter = list(minlen = 2, supp = 0.005, conf = 0 ...

```
> inspect(rules.sorted)
```

	lhs	rhs	support	confidence	coverage	lift	count
[1]	{Class=2nd, Age=Child}	=> {Survived=Yes}	0.011	1.000	0.011	3.096	24
[2]	{Class=2nd, Sex=Female, Age=Child}	=> {Survived=Yes}	0.006	1.000	0.006	3.096	13
[3]	{Class=1st, Sex=Female}	=> {Survived=Yes}	0.064	0.972	0.066	3.010	141
[4]	{Class=1st, Sex=Female, Age=Adult}	=> {Survived=Yes}	0.064	0.972	0.065	3.010	140
[5]	{Class=2nd, Sex=Female}	=> {Survived=Yes}	0.042	0.877	0.048	2.716	93
[6]	{Class=Crew, Sex=Female}	=> {Survived=Yes}	0.009	0.870	0.010	2.692	20
[7]	{Class=Crew, Sex=Female, Age=Adult}	=> {Survived=Yes}	0.009	0.870	0.010	2.692	20
[8]	{Class=2nd, Sex=Female, Age=Adult}	=> {Survived=Yes}	0.036	0.860	0.042	2.663	80
[9]	{Class=2nd, Sex=Male, Age=Adult}	=> {Survived=No}	0.070	0.917	0.076	1.354	154
[10]	{Class=2nd, Sex=Male}	=> {Survived=No}	0.070	0.860	0.081	1.271	154
[11]	{Class=3rd, Sex=Male, Age=Adult}	=> {Survived=No}	0.176	0.838	0.210	1.237	387
[12]	{Class=3rd, Sex=Male}	=> {Survived=No}	0.192	0.827	0.232	1.222	422

Conclusion

This lab demonstrated the complete Apriori association rule mining pipeline in R. After transforming the aggregated Titanic table into an individual-level transaction dataset of 2,201 records, we extracted 27 rules under default thresholds and then refined the search to 12 survival-targeted rules sorted by lift. The results reveal strong patterns: second-class children had a 100% survival rate, while third-class adult males showed the highest probability of not surviving — findings consistent with historical records of the disaster.